

The KDD Process

Comprehensive Study Notes

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1. The Context: From Data to Knowledge Society

We are transitioning from an industrial society (key resources: energy & capital) to a post-industrial one, where information and knowledge are the key strategic assets. Proper information management can ensure an organization's survival and competitive differentiation.

1.1 Industrial Revolutions — Quick Reference

Revolution	Key Technologies
1st	Mechanization, water & steam power
2nd	Mass production, assembly line, electricity
3rd	Computers & automation
4th	Cyber-physical systems, AI, IoT

1.2 Key Concepts

- Information Society — knowledge as the central economic resource
- Big Data — massive, heterogeneous, fast-moving data streams
- Data Science — cross-disciplinary analysis of data
- Knowledge Worker — professional whose main resource is knowledge

2. Big Data

2.1 Scale: What is a Zettabyte?

Unit	Value
1 Zettabyte	10^{21} bytes
	10^{15} Megabytes (MB)
	10^{12} Gigabytes (GB)
	10^9 Terabytes (TB) — one billion terabytes

2.2 The Five V's of Big Data

V	Definition	Key Point
Volume	Enormous amount of data generated daily	Sensors, social media, financial transactions
Velocity	Speed of data generation and processing	Real-time data requires timely analysis
Variety	Multiple data types and sources	Structured (DB tables), semi-structured (JSON), unstructured (images, audio, text)

Veracity	Quality and reliability of data	Data may contain errors, duplicates, or be incomplete
Value	Ability to extract useful insights	Not all data is equally useful; meaningful patterns support strategic decisions

2.3 Data-Driven Innovation

The EU Commission promotes data-driven innovation: using computational data analysis to develop goods and services that improve individual and organizational daily life.

Data analytics applied to Big Data enables organizations to:

- Personalize products/services
- Optimize processes
- Forecast market trends
- Improve customer experience
- Reduce decision-making risks

3. Data, Information & Knowledge

3.1 Definitions

Concept	Definition	Key Property
Data	Raw, uninterpreted facts recorded in some medium	No meaning in isolation → Uncertainty
Information	Result of processing/interpreting data; reduces uncertainty	Has meaning in a context → Certainty
Knowledge	Understanding gained from experience/study; ability to use information meaningfully	Enables action and decisions

3.2 Example

Data: 30°C

Information: The temperature in Rome is 30°C

Knowledge: With 30°C in Rome in summer, wear light clothing and protect yourself from the sun.

3.3 The DIKW Pyramid

The knowledge hierarchy (bottom to top):

- Data — raw facts, no meaning
- Information — data in context
- Knowledge — actionable understanding
- Wisdom — judicious application of knowledge (extended pyramid)

 *Key distinction: data can be collected automatically; knowledge requires experts and involves abstractions that cannot be objectively verified.*

4. Data Science & Data Lifecycle

4.1 Data Science

Data science is the art and science of analyzing structured and unstructured data to identify patterns, make predictions, and take informed data-driven decisions. It sits at the intersection of Computer Science, Mathematics/Statistics, and Business/Domain Expertise.

4.2 The 6-Phase Data Lifecycle

Phase	Description	E-commerce Example
1. Generation	Data is produced through events or actions	User clicks, cart additions, purchases, reviews
2. Acquisition & Recording	Data is tracked and stored	Cookies, DB purchase records, session logs, GDPR consent
3. Processing & Management	Data is cleaned and prepared	Removing duplicates, standardizing formats, integrating web+mobile
4. Analysis	Patterns are extracted	Product predictions, cart abandonment, customer segmentation, seasonal trends
5. Visualization & Interpretation	Insights are communicated	Sales charts, heatmaps, automated reports, real-time KPIs
6. Storage or Deletion	Data is archived or erased	Tax archiving, inactive user deletion, GDPR requests, encrypted backups

5. Knowledge Discovery in Databases (KDD)

5.1 Origins

- 1989: Gregory Piatetsky-Shapiro organized the first KDD workshop at IJCAI
- Goal: explore how Machine Learning (ML) can extract knowledge from large datasets
- 1995 (First KDD Conference, Montreal): KDD officially denotes the whole process from data collection to results interpretation
- Data Mining = the step within KDD where algorithms actually extract information from data

5.2 Definition of Knowledge Discovery

"Knowledge discovery is the nontrivial extraction of implicit, previously unknown, and potentially useful information from data."

— Fayyad, Piatetsky-Shapiro & Smyth, 1996

Breaking down the definition:

Term	Meaning
Nontrivial	Involves search or inference; not just computing a predefined quantity (e.g., an average)
Implicit	Information is hidden in the data, not explicitly available (explicit info → database querying)
Previously unknown	Must be novel; depends on the frame of reference (e.g., a rule trivial for humans may be new for a machine)
Potentially useful	Must help achieve a goal; patterns unrelated to goals are not knowledge

5.3 Formal Definition: Pattern

Given a set of facts F , a representation language L , and a certainty measure C , a pattern is a statement S in L describing relationships among a subset of F with certainty c , such that S is simpler than enumerating all facts.

A pattern is knowledge if it is interesting and sufficiently certain (valid).

Certainty factors:

- Integrity of the data
- Size of the sample used for discovery
- Support from available domain knowledge

Interestingness criteria:

- Novel
- Useful
- Nontrivial to compute

5.4 Patterns vs. Models

Concept	Description	Example
Pattern	Local feature of the dataset (limited to a subset of observations/attributes)	$f(x) = 3x^2 + x$
Model	Global summary of the dataset; fitted models play the role of inferred knowledge	$f(x) = \alpha x^2 + \beta x$

6. The KDD Process

6.1 Overview

KDD is an interactive and iterative process involving numerous steps, with many decisions made by the user. It is not a linear pipeline — iterations back to previous steps are common.

6.2 KDD Steps (Fayyad et al.)

- Understanding the application domain, background knowledge, and end-user goals
- Creating a target dataset (selecting relevant variables and data samples)
- Data cleaning & preprocessing: noise removal, handling missing values, time sequence info
- Data reduction & projection: dimensionality reduction, finding useful features
- Choosing the data mining task (classification, regression, clustering, etc.)
- Choosing the data mining algorithm(s) and parameters
- Data mining: searching for patterns (classification rules, regression, clustering, etc.)
- Interpreting mined patterns; possibly returning to earlier steps
- Consolidating/documenting discovered knowledge; resolving conflicts with prior knowledge

📌 *Most research focuses on Step 7 (data mining), but all steps are critical in practice.*

7. CRISP-DM — The Industry Standard Process

7.1 Background

- Full name: Cross Industry Standard Process for Data Mining
- Founded by: DaimlerChrysler, SPSS, NCR — CRISP 1.0 released 1999
- Goal: industry, tool, and application-neutral process for knowledge discovery
- The phase sequence is not strict — movement back and forth between phases is expected
- The process is cyclic: a deployed solution can feed a new KDD cycle

7.2 The Six CRISP-DM Phases

Phase	Key Tasks	Outputs
1. Business Understanding	Determine business objectives; assess situation; determine data mining goals; produce project plan	Business objectives, success criteria, project plan, cost-benefit analysis
2. Data Understanding	Collect initial data; describe data; explore data; verify data quality	Initial data collection report, data description report, data exploration report, data quality report
3. Data Preparation	Select data; clean data; construct data (derived attributes); integrate data; format data	Dataset, data cleaning report, derived attributes, merged/reformatted data
4. Modeling	Select modeling technique; generate test design; build model; assess model	Modeling technique, test design, parameter settings, model description, model assessment
5. Evaluation	Evaluate results vs. business success criteria; review process; determine next steps	Assessment of data mining results, approved models, list of possible actions, decision
6. Deployment	Plan deployment; plan monitoring & maintenance; produce final report; review project experience	Deployment plan, monitoring plan, final report, documentation

8. Business Understanding (Detail)

8.1 Determine Business Objectives

Minimum requirements: a perceived business problem or opportunity, and some level of executive sponsorship. Requires collaboration between business analyst and data analyst.

Important: set realistic expectations early. E.g., improving mailing campaign response from 1% to 2–2.5% is realistic; dramatically higher targets will likely doom the project.

8.2 Data Mining Goals: Directed vs. Undirected

Type	Also Known As	Tasks
Directed / Predictive	Supervised Learning	Classification, Estimation, Prediction
Undirected / Descriptive	Unsupervised Learning	Affinity grouping, Clustering, Description & Profiling

9. Data Mining Tasks

9.1 Classification

Learning a function that maps a data item to one of several predefined classes.

Example: Given a person's income and debt, classify whether they will default on a loan (x) or not (o). The algorithm finds a boundary dividing the feature space into class regions.

9.2 Regression

Learning a function mapping a data item to a real-valued prediction variable.

Example: Fitting total debt as a linear function of income; predicting consumer demand as a function of advertising expenditure.

9.3 Affinity Grouping / Association Rules

Finding associations between groups of variables.

Association rules: $X \rightarrow Y$ (transactions containing X tend to contain Y).

Example: "98% of customers who purchase tires and automotive accessories also have automotive services carried out."

9.4 Clustering

Partitioning data into homogeneous groups without predefined labels (unsupervised).

Related task: probability density estimation — estimating the joint multivariate probability density function of all variables.

Examples: customer segmentation in marketing, identification of spectral sub-categories in astronomy.

9.5 Summarization / Description / Profiling

Finding a compact description for a subset of data (e.g., computing means and standard deviations across variables). Typical of periodic reporting.

9.6 Change & Deviation Detection

Discovering significant changes from previously measured or normative values.

- Outlier detection — identifying 'strange examples' possibly due to measurement errors
- Emerging patterns — patterns frequent in one class but rare in others; characterize a class

Applications: fraud detection (credit cards, insurance, telephone), quality control, defect tracing, loan analysis.

10. Case Study: Customer Vulnerability Analysis (CVA)

Business Objective

Maintain market share for a particular brand of frozen orange juice (12-oz. can) in the Midwestern U.S. during November–January, using a direct mail campaign targeting vulnerable customers.

Key Question

Which households should be included in the direct mail campaign for it to be effective?

KDD Strategy

- Select: customers who purchase 12-oz. frozen OJ from the data warehouse
- Transform: classify customers as Loyal (buys same brand $\geq 80\%$ of time) based on average price/oz
- Assimilate: identify vulnerable customers using behavioral models
- Model building: use syndicated data to build models of typical buying patterns for both vulnerable and loyal customers
- Segmentation: divide customer database into homogeneous groups for targeted marketing campaigns

11. Quick Reference — Key Terms

Term	One-line Definition
KDD	The complete process of extracting knowledge from data (collection → interpretation)
Data Mining	Step 7 of KDD: applying algorithms to extract patterns from data
CRISP-DM	Industry-standard cyclic 6-phase model for KDD projects
Supervised Learning	Learning with labeled examples (classification, regression)

Unsupervised Learning	Learning without labels (clustering, association rules)
Pattern	A statement describing relationships in data, simpler than enumerating all facts
Model	A global summary of the dataset; fitted parameters = inferred knowledge
Interestingness	A pattern is interesting if it is novel, useful, and nontrivial to compute
Veracity (Big Data)	Quality and reliability of data; errors negatively impact analyses
Data Lifecycle	6 phases: Generation → Acquisition → Processing → Analysis → Visualization → Storage/Deletion

12. CRISP-DM Phase 2: Data Understanding

Data Understanding covers all activities needed to become familiar with the data before any preparation or modeling begins. It has four main tasks.

12.1 Collect Initial Data

Access relevant data from the inventory of resources (e.g., load from a data warehouse).

Output: Initial data collection report — lists data locations, acquisition methods, and any problems encountered.

12.2 Describe Data

First examination of the properties of the data.

Output: Data description report — describes format, potential values, quantity, field identities, and any notable characteristics. Metadata from a data warehouse can help.

Variable Types

Type	Subtype	Description	Examples
Categorical	Nominal	Finite values, no order among them	Gender (M/F), Marital status (single/married/divorced), Education level
Categorical	Ordinal	Finite values with a meaningful order	Customer credit rating (good/regular/poor)
Quantitative	Discrete	Integer values, arithmetic allowed	Number of employees, Month of year
Quantitative	Continuous	Real-number values, arithmetic allowed	Income, Average purchases, Revenue, Temperature

12.3 Explore Data

Statistical methods and visualization techniques are applied to understand the data distribution and spot problems.

Output: Data exploration report.

Exploration tools by variable type:

Variable Type	Technique	Purpose
Categorical	Frequency distributions, histograms, pie charts	Identify distribution skews, invalid or missing values
Quantitative	Max, min, mean, mode, median, standard deviation	Detect invalid/skewed data; extreme values reveal spurious entries
Any	Scatterplots	Show relationships between two or more continuous variables
Any	Boxplots	Compare center (median) and spread (IQR) of two or more variables; identify outliers

12.4 Verify Data Quality

Four dimensions of data quality:

Dimension	Definition
Accuracy	Conformity of stored value to the actual value
Completeness	No missing values
Consistency	Uniform representation across all records
Up-to-dateness	Stored data is not obsolete

Important considerations:

- Most operational data was never designed for data mining — it comes from inconsistent, poorly documented systems
- External/purchased databases may have up to 400 variables, sparsely filled and potentially outdated
- Time sensitivity is critical: e.g., customers change jobs every year — analyses using job type must be re-examined periodically

Output: Data quality report — documents findings and proposes solutions if quality problems exist.

12.5 CVA Case Study — Data Understanding

Main data source: a syndicated database of ~15,000 consumers whose supermarket purchases were tracked for 3 years via fidelity cards.

Data recorded per supermarket visit: Customer ID, Date of visit, Basket (list of items, quantities, promotion IDs), Basket value.

Database schema (3 tables):

Table	Fields	Variable Types
Visits	Visit ID (PK), Customer ID (FK), Date of visit, Basket value, Payment mode	Nominal, Nominal, Date, Continuous, Nominal
Purchases	Purchase ID (PK), Visit ID (FK), Product ID (FK), Product quantity, Promotion ID	Nominal, Nominal, Nominal, Discrete, Nominal

Products	Product ID (PK), Description, Quantity, Brand	Nominal, Nominal (free text), Discrete/Continuous, Nominal
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 *Data quality issue found: some records in Purchases have Product IDs that do not reference a valid product in Products — a partial basket view is built for these cases.*

13. CRISP-DM Phase 3: Data Preparation

Data Preparation covers all activities to transform raw data into an analytical dataset ready for modeling. It consists of five tasks: Select Data, Clean Data, Construct Data, Integrate Data, and Format Data.

13.1 Select Data

Choose which tuples and attributes to include, based on:

- Relevance to the data mining goals
- Data quality and technical constraints
- Limits on data volumes or data types

Output: Report with rationale for inclusion/exclusion of data.

Selection can be performed manually or automatically (via sampling and/or feature selection).

13.1.1 Sampling

Sampling reduces dataset size while preserving representativeness. The larger the sample, the more likely it reflects the population.

Sampling Method	Description	Key Notes
Simple Random (with/without replacement)	Every possible group of the required size has equal selection probability	Most straightforward; works well when population is large vs. sample
Stratified Random	Population divided into strata; a random sample is drawn from each stratum separately	Ensures underrepresented groups are included; adjust weights when aggregating across strata
Cluster Sampling	Natural clusters (e.g., geographic areas, time periods) are randomly selected; all members of chosen clusters are used	Economizes travel/access cost; block sampling in databases reads full data blocks
Two-Stage Sampling	First randomly select clusters, then sample within each cluster	Combines benefits of cluster and random sampling
Systematic (every k-th)	Number population, pick a random start between 1 and k, then take every k-th element	Easy to implement even when population size is unknown; beware periodicities in the list

Stratified sampling — key rules:

- Strata should have means that differ substantially from each other
- Minimize within-stratum variance, maximize between-stratum variance
- Sample size proportional to stratum standard deviation (unless a minimum count per stratum is needed)

⚠ *When strata are over-sampled (to meet a minimum count), correct for this by weighting stratum averages by their actual proportion in the population when computing overall statistics.*

How large should the sample be?

- Use power analysis (statistics): estimates sample size needed to detect a specified effect size with a chosen probability
- Progressive/adaptive/sequential sampling: keep growing the sample until results stabilize

13.1.2 Feature Selection

Real-life databases can contain many attributes (features). Feature selection reduces dimensionality by keeping only the most informative subset.

The Curse of Dimensionality

As the number of features (dimensions) increases:

- The required sample size grows exponentially to maintain the same accuracy
- With a fixed number of training examples, classifier performance typically degrades
- Not enough observations to estimate distributions reliably

Three approaches to feature selection:

Approach	How it works	Pros	Cons
Wrapper Model	Uses a DM algorithm itself to score feature subsets. Three steps: (1) generate candidate subsets, (2) train a model on each, (3) pick the best-performing subset.	Optimized for the specific algorithm and problem	Computationally expensive (cross-validation at each iteration)
Filter Model	Evaluates feature subsets using intrinsic data properties (independent of the DM algorithm). Measures: Information Gain, Distance measures.	Fast; algorithm-agnostic	May be suboptimal for a specific learner
Embedded Methods	Feature selection is an integral part of the learning algorithm itself (e.g., decision trees).	No separate selection step needed	Cannot be decoupled from the learning algorithm

Subset generation strategies (Wrapper & Filter):

- Exhaustive enumeration — try all 2^N subsets; guaranteed optimal but infeasible for large N
- Random generation — randomly sample subsets
- Forward selection — start empty, add one feature at a time (greedy)
- Backward elimination — start full, remove one feature at a time (greedy)

Model evaluation methods (Wrapper):

Method	Description	Trade-off
Holdout	Split data into training set and test set; train on train, evaluate on test	Easy to implement; may give biased estimate if split is unlucky
k-fold Cross-Validation	Split into k folds; train on k-1 folds, test on the remaining fold, repeat k times, average results (k = 5 or 10 typical)	More reliable estimate; computationally heavier

 *Embedded methods (e.g., decision trees) are not considered here since they are part of the modeling step, not a separate data preparation step.*

13.2 Clean Data

Raises data quality to the required level based on the data quality report. Involves selecting clean subsets, inserting defaults, or estimating missing values.

Output: Data cleaning report — describes decisions/actions, lists transformations applied, and notes possible impacts on results.

Three main data quality issues:

Issue	Definition	Treatment
Noisy Data	Observed values differ from actual values (human error or latent factors). E.g., age = 650; negative income.	Correct if a valid substitute can be found, otherwise drop the record
Outliers / Anomalies	Values that are rare but possible; skewed distributions often indicate them. Distinguished from noise by whether they are genuine (real high earners) or errors.	Context-dependent: keep if genuine, remove/correct if due to poor collection
Missing Values	Simply absent, or invalid values deleted during noise detection. Causes: human error, data not available at input time, heterogeneous source mismatches.	See techniques below

Techniques for missing values:

Technique	Description	Drawback
Drop the observation	Delete any row that has at least one missing value	Loss of potentially valuable data
Drop the variable	If a large proportion of observations is missing for the same variable, remove it from the analysis	Loss of the entire attribute
Replace with central value	Quantitative: use mean or median. Categorical: use mode or a new 'unknown' category.	Introduces artificial data; may bias distributions
Predictive imputation	Use a model (e.g., regression) to predict the missing value from the other attributes	More sophisticated; relies on model accuracy

13.3 Construct Data

Generate derived variables, new records, or transformed versions of existing variables to better suit the data mining algorithms.

Output: Derived variables, generated records, and single-variable transformed data.

Single-variable transformations (examples):

- Converting date from US/European to Julian format
- Computing age from date of birth
- Aggregating average account balance over 3, 6, and 12-month periods

Scaling / Normalization

Normalization transforms numerical columns to a new range, ensuring equal treatment of all variables and meeting input requirements of some algorithms.

Method	Formula	When to use	Notes
Min-Max Normalization	$v' = (v - \min A) / (\max A - \min A) \times (\text{new_maxA} - \text{new_minA}) + \text{new_minA}$	Preserves original relationships; common target [0,1]	Out-of-bounds error if future values fall outside the training range
Z-score Standardization	$v' = (v - \text{meanA}) / \text{stddev_A}$	When min/max are unknown, or outliers dominate	Result has mean = 0, std = 1 (unit variance)
Decimal Scaling	$v' = v / 10^j$ (where j = smallest int so that $\max v' < 1$)	Simple rescaling to (-1, +1)	Number of decimal places moved depends on max absolute value

Discretization (binning)

Converts quantitative variables into categorical ones by dividing values into intervals (bins). Used when a DM algorithm requires categorical input.

Method	Rule	Pros	Cons	Example (values: 0,4,12,16,16,18,24,26,28)
Equal-Width	$W = (\max - \min) / N$; all bins have the same width	Straightforward	Sensitive to outliers; skewed data handled poorly	$N=3 \rightarrow [-,10): 0,4 \mid [10,20): 12,16,16,18 \mid [20,+): 24,26,28$
Equal-Depth (Frequency)	Each bin contains approximately the same number of samples	Good data scaling; minimizes info loss	Interpreting categorical attributes can be tricky	$N=3 \rightarrow [-,14): 0,4,12 \mid [14,21): 16,16,18 \mid [21,+): 24,26,28$

Rule of thumb for number of bins: $N_{\text{bins}} = M / (3 \times C)$, where M = number of training examples and C = number of classes.

One-of-N Encoding (One-Hot Vector)

Converts a categorical variable to a binary numeric representation. Special case: one-hot vector — a binary vector with exactly one 1.

Example: Car type = {Ford, Lincoln, Nissan} → Ford = 100, Lincoln = 010, Nissan = 001.

 *One-hot encoding is typically required before feeding categorical data into neural networks.*

Dimensionality Reduction: Factor Analysis

Factor Analysis (and Principal Component Analysis) build a small set of indices (factors) from a large number of variables, reducing the feature space while minimizing information loss.

Concept	Explanation
Factor	A latent dimension representing a cluster of variables that correlate strongly with each other
Factor Loading	Correlation between an original variable and a factor; ranges from -1.00 to +1.00
Factor Score	Value computed for each observation: $Y = X_1 \cdot (\text{var1}) + X_2 \cdot (\text{var2}) + \dots$ (weighted sum of input variable scores)
Algebraic form	$Y = X_1 \cdot (\text{variable 1}) + X_2 \cdot (\text{variable 2}) + \dots$ where X_1, X_2 = factor loadings
Loading interpretation	$\geq \pm 0.30$ = minimal; $\geq \pm 0.40$ = more important; $\geq \pm 0.50$ = practically significant

Factor Analysis vs. Cluster Analysis:

- Factor analysis: clusters variables (reduces the number of features)
- Cluster analysis: clusters observations (groups rows/records)

Reducing input variables:

- Advantage: smaller, more manageable feature set for further analysis
- Disadvantages: difficult to perform; information loss; results harder to interpret

13.4 Integrate Data

Combines information from multiple datasets/tables/records into a unified analytical data model — a consolidated, integrated, time-dependent restructuring ready for mining.

Output: Merged data (joining datasets about the same objects).

Unit of Analysis vs. Unit of Observation

Concept	Definition	Example
Unit of Observation	The entity on which data are directly collected	Individual pupils in a school study
Unit of Analysis	The entity about which conclusions are drawn	The class (aggregated from individual pupils)

Key rule	You can aggregate upward (observation → analysis); you cannot disaggregate downward	Cannot infer individual behavior from aggregate data
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⚠ *Ecological fallacy (ecological inference): drawing conclusions about individual behavior from aggregate data. If you want conclusions about persons, persons must be the unit of analysis.*

The Single-Table Assumption

Most DM algorithms operate on a single two-way table (propositional learning): rows = units of analysis, columns = features. When the data naturally spans multiple relations, you must decide how to flatten it:

Strategy	Description	Problem
One row per order (denormalized)	Each order gets its own row in the customer table	Analysis is about orders, not customers. Redundancy and anomalies from non-normalized data.
Aggregate into one row per customer	Summarize all orders into a single customer record (e.g., total orders, number of stores)	Much less information; interactions between attributes (e.g., payment mode × store type) are lost

Multi-Relational Data Mining (MRDM)

Algorithms that can analyze data distributed across multiple relations, as found in relational databases. They come from Inductive Logic Programming (ILP).

- ILP originated in automated program synthesis from examples
- Scope has broadened to all DM tasks: association rules, regression, clustering, etc.
- Patterns expressed in richer languages: predicate calculus (first-order logic), SQL, graphs
- Types: relational classification rules, relational regression trees, relational association rules

Example relational classification rule:

*IF Customer(C1,...,Income1,...) AND Order(C1,O1,...,Pay1)
AND Pay1 = credit_card AND Income1 > 108000
THEN Response = Yes*

Householding (CRM application)

A data integration technique that consolidates dispersed customer records to build a profile of an organization's relationship with a household. The consolidated profile includes all household members and all products/services they use.

13.5 Format Data

Syntactic modifications that do not change meaning but are required by the modeling tool (e.g., reordering records, rearranging attributes).

Output: Reformatted data.

Common data file formats:

Format	Description	Key Features
CSV (Comma Separated Values)	Records as rows, attributes separated by commas. Supported by virtually all spreadsheet and database tools.	Universal; no schema; no type information
ARFF (Attribute-Relation File Format)	Proprietary format of Weka. Includes a header section (@relation, @attribute) that declares attribute names and types, followed by @data.	Supports categorical, numeric, string, and date attributes; supports sparse data; type-safe
Sparse ARFF	Extension of ARFF for datasets where most values are zero (e.g., word counts in text categorization). Non-zero values encoded as {index value} pairs.	Efficient storage; zero values omitted; missing values must be explicit (?)

ARFF data types:

- Nominal — finite set of values listed in curly braces: @attribute outlook {sunny, overcast, rainy}
- Numeric — continuous or discrete number
- String — like nominal but the list of values is not pre-specified (treated as nominal internally)
- Date — ISO-8601 format: yyyy-MM-ddTHH:mm:ss (treated as numeric internally)

14. CVA Case Study — Data Preparation

14.1 Feature Engineering: 20 Derived Variables

The team collapsed 3 years of purchase history per customer into a single record by creating 20 new variables (unit of analysis = customer). Key examples:

Variable	Definition
Primary-Brand	The OJ brand of which the household bought the most ounces over 3 years
Purchase-Frequency	Ratio of number of purchases to number of days between first and last supermarket visit
Ounces-Per-Day	Ratio of total ounces of OJ purchased to number of days between first and last visit

14.2 Labelling: Loyal vs. Vulnerable

Target variable definition:

- Loyal customer: bought the same OJ brand more than 80% of the time
- Vulnerable customer: all others (bought varying brands)

The resulting OJ database: 1,986 records, each labelled loyal or vulnerable.

 *This labelling transforms the problem into a supervised classification task: predict whether a customer is vulnerable based on their purchase behavior features.*

15. Quick Reference — Key Terms (Part 2)

Term	One-line Definition
Nominal variable	Categorical variable with no order among values (e.g., gender, marital status)
Ordinal variable	Categorical variable with a meaningful order (e.g., credit rating: good > regular > poor)
Discrete variable	Quantitative variable with integer values (e.g., count of employees)
Continuous variable	Quantitative variable with real-number values (e.g., income, temperature)
Curse of dimensionality	Required sample size grows exponentially with number of features; classifier performance degrades
Wrapper model	Feature selection using the DM algorithm's own performance as the evaluation criterion
Filter model	Feature selection using intrinsic data measures (e.g., information gain), independent of the DM algorithm
Min-Max normalization	Linear scaling to [0,1] (or any [new_min, new_max]); preserves relationships but sensitive to outliers
Z-score standardization	Rescales to mean=0, std=1; robust to unknown min/max and outliers
Equal-width binning	Discretization with bins of equal interval width; sensitive to skew and outliers
Equal-depth binning	Discretization where each bin has approximately the same number of samples
One-hot encoding	Converts a categorical variable to a binary vector with a single 1; required by many neural networks
Factor analysis	Reduces many variables to fewer latent factors by grouping correlated variables; factor loadings $\geq \pm 0.50$ = practically significant
Ecological fallacy	Error of inferring individual behavior from aggregate-level data
Single-table assumption	Most DM algorithms require data in one flat table; multi-relational data must be flattened or MRDM algorithms used
ARFF	Weka's typed data format; supports nominal, numeric, string, date, and sparse representations
Householding	CRM technique consolidating dispersed customer records into a unified household profile

16. CRISP-DM Phase 4: Modelling

The Modelling phase is where actual data mining techniques are selected and applied to the prepared data. It has four main tasks and connects closely to Data Preparation — results here may require cycling back.

16.1 Select Modeling Technique

The analyst chooses the specific algorithm(s) based on:

- Nature of the objective (classification, regression, clustering, etc.)
- Ability to handle certain data types and multiple relations
- Explicability of the output (interpretable vs. black-box)
- Scalability and level of familiarity

Output: The selected modeling technique and any assumptions it requires (e.g., probability distribution, no missing values, nominal class attribute).

Any data mining algorithm has three primary components:

Component	Description	Examples
Model Representation	Language L used to describe discoverable patterns	Decision trees, logic rules, visual plots
Model Evaluation	Estimates uncertainty of a pattern	Cross-validation, accuracy, F1-score
Search	Strategy for exploring the hypothesis space	Exhaustive, greedy

16.2 Model Representation

Model representation is the language used to describe discoverable patterns. It depends on the type of representation, the descriptive capacity, and appropriateness for the intended user.

Pattern Types

- Interfield patterns: relate values of fields in the same record (e.g., Procedure = surgery $\Rightarrow$ days-in-hospital > 5)
- Interfield-record patterns: relate values aggregated over groups of records (e.g., diabetic patients have twice as many complications as non-diabetics)

Quantitative vs. Qualitative Discovery

- Quantitative: numeric relationships using mathematical equations (e.g., $X^2 + 0.5Y = 2.65$)
- Qualitative: logical relationships among fields (e.g., Procedure = surgery $\Rightarrow$ days-in-hospital > 5)

Multiple simple implications can be combined into a logical theory, which under some assumptions can be treated as a causal theory.

Representation Forms by User Type

User Type	Representation Form
Humans	Natural language, logical rules (IF...THEN), decision trees, visual plots
Computer programs / Expert systems	Programming languages, declarative formalisms (e.g., Prolog)
The discovery system itself	Domain knowledge and discovered knowledge share a common representation

⚠ *If representation is too limited, no amount of training will produce an accurate model.*

⚠ *A more powerful representation increases the risk of overfitting — resulting in reduced accuracy on unseen data.*

Representing Uncertainty

Patterns are often associated with degrees of uncertainty. Methods include: Probability, Standard deviation, Fuzzy sets.

16.3 Model Evaluation (within Modelling)

Model evaluation estimates the uncertainty of a pattern. Cross-validation is one key method.

Cross-Validation Procedure (k-fold)

- Shuffle all items in the dataset
- Divide into k equal subsets of size n
- For each i from 1 to k: use subset i as test set, train on the remaining k-1 subsets, record performance
- Clear memory (forget learned knowledge)
- Calculate average performance over all k test sets

16.4 Generate Test Design

Before building models, define a procedure to test model quality.

Output: A test design describing the plan for training, testing, and evaluating models. For classification tasks, error rates are commonly used as quality measures; data is split into training and testing sets.

16.5 Build Model

The chosen data mining technique is applied to the prepared data.

Output: Parameter settings with rationale, output models with expected accuracy, robustness, and potential shortcomings.

Output in PMML (Predictive Model Markup Language)

PMML is an XML-based standard for sharing predictive models across tools and vendors (IBM, Oracle, SAS, Microsoft, SPSS, etc.).

- Goal: define and share predictive models using an open, vendor-independent standard
- Latest version: PMML 4.4 (approved November 2019)

Supported model types: Tree Model, Neural Networks, Clustering, Regression, General Regression, Naïve Bayes, Association Rules, Sequence Rules, Text Model, Rule Set, Support Vector Machine.

16.6 Assess Model

Modeling results are interpreted against data mining success criteria. Key questions: How accurate is the model? How well does it describe observed data? How comprehensible is it? Output: Assessment of generated models and revision of parameter settings.

Assessing Descriptive Models — Minimum Description Length (MDL)

The MDL principle measures the number of bits needed to encode both the rule and all exceptions to it. Fewer bits = better, more expressive rule.

Assessing Predictive Models

- Evaluated on accuracy on previously unseen data (test set must be independent from training set)
- Two models with the same overall accuracy may have very different variance among individual predictions
- When errors have different costs, an error cost measure is more appropriate than misclassification rate

Confusion Matrix

A confusion matrix is used to assess the performance of a classifier:

	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TP)	False Positives (FP)
Predicted Negative (0)	False Negatives (FN)	True Negatives (TN)

Classification Metrics

Metric	Formula	When to use
Accuracy	$A = (TP + TN) / \text{Total}$	General correctness; sensitive to class imbalance
Precision	$P = TP / (TP + FP)$	When false positives are costly (e.g., spam filtering, medical diagnosis)
Recall	$R = TP / (TP + FN)$	When false negatives are costly (e.g., fraud detection, defect detection)
F1-Score	$F1 = 2 \times (P \times R) / (P + R)$	When you need a balance between precision and recall

Precision vs. Recall tradeoff: increasing one reduces the other. The F1-Score uses the harmonic mean (instead of arithmetic mean) to penalize extreme values more effectively.

Worked Example — Confusion Matrix

Given: TP = 100, FN = 5, FP = 10, TN = 50 | Total = 165 predictions

Accuracy = $(100 + 50) / 165 = 0.90$

Precision = $100 / (100 + 10) = 100 / 110 = 0.91$

Recall = $100 / (100 + 5) = 100 / 105 = 0.95$

F1-Score = $2 \times (0.91 \times 0.95) / (0.91 + 0.95) = 0.9296$

Regression Models — MSE

For regression models, accuracy is estimated using Mean Square Error (MSE) — the average of squared differences between predicted and actual values:

$$\text{MSE} = (1/n) \times \sum (y_i - \hat{y}_i)^2$$

17. CRISP-DM Phase 5: Evaluation

The Evaluation phase assesses models not just on technical accuracy, but against the original business objectives. It has two main tasks.

17.1 Evaluate Results

Whereas model assessment focuses on accuracy and generality, this task evaluates models according to the original business objectives and success criteria.

- Requires collaboration between a skilled data analyst and a business analyst
- The business analyst formulates findings in terms directly relevant to the original business objectives
- A data analyst with domain/industry knowledge is a great asset at this stage

Output: Overall assessment of data mining results with respect to business success criteria, including a final statement on whether the project meets its initial objectives.

An important benefit of a project may simply be a renewed organizational focus on the corporate data asset and the power of data-driven solutions — promoting longer-term initiatives such as data warehousing.

17.2 Review Process

In addition to model evaluation, the entire CRISP-DM process is reviewed to identify any neglected factors or to establish a reusable procedure for similar future models.

Key question: "Have we found something that is interesting, valid, and operational?" — Almost always the answer is no, leading to another iteration of the exploratory cycle.

Output: Recommendations for further activities.

18. CRISP-DM Phase 6: Deployment

The Deployment phase puts the results of data mining into action. It involves translating new, valid, operational knowledge into concrete business changes.

18.1 Plan Deployment

A deployment strategy is developed to integrate data mining results into the business.

- New predictive models and association rules may be included in existing application code, expert system shells, or database procedures
- The specific actions depend on the type of application and executive commitments made at the project's outset

Output: A deployment plan.

Common deployment approaches (from survey data):

Deployment Method	Usage %
Deploy in batch mode	50.5%
Use data mining tool for scoring	49.5%
Deploy in production	48.4%
Use findings to change business rules	44.2%
Publish research papers	38.9%
Deploy in real-time mode	22.1%
Convert model to SQL	21.1%
Convert model to another language / C / Java	16.8%
Convert model to PMML	4.2%

18.2 Plan Monitoring and Maintenance

Careful preparation of a maintenance strategy avoids unnecessarily long periods of incorrect usage of data mining results.

Output: A maintenance plan covering the monitoring process.

18.3 Produce Final Report

At the end of the project a final report should be produced:

- A summary of the project
- A final presentation of the data mining results

Output: A report or final presentation (or both).

18.4 Review Project

Assess what went right and what went wrong; what was done well and what needs improvement.

Output: Experience documentation to consolidate lessons for future engagements, including: pitfalls, misleading approaches, and hints for selecting the best-suited data mining technique in similar situations.